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# Accelerating perovskite solar cell performance prediction with machine learning on existing experimental data

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ABSTRACT

Herein, we proposed a machine learning (ML) framework for determining four photovoltaic parameters: open-circuit voltage ($V_{oc}$), short-circuit current density ($J_{sc}$), fill factor (FF), and power conversion efficiency (PCE) in perovskite solar cells. Initially, a dataset of 503 experimental entries was manually compiled from different peer-reviewed literatures, and physically meaningful descriptors like perovskite composition, energy levels of the various layers, band offsets, charge mobilities, and the architecture type were used as inputs for the ML models. The four ensemble learning algorithms — Random Forest, Gradient Boosting, Extreme Gradient Boosting, and CatBoost — were then trained and analyzed with 5-fold cross-validation and the reserved test sets. CatBoost showed the highest accuracy for PCE prediction (RMSE = 1.09%, $r$ = 0.964), while Random Forest performs best for $V_{oc}$, $J_{sc}$, and FF, achieving RMSE as low as 0.033 V, 1.013 mA/cm$^2$, and 0.031, respectively. SHAP-based interpretability analysis revealed that intrinsic features of perovskite, charge mobilities, and interfacial band alignment are essential to the device performance. Furthermore, SHAP feature dependency plots were also used to study how the individual features are associated with the target's prediction. Finally, an additional evaluation based on 12 independent samples not part of training or test sets confirmed model robustness with predictions in close agreement with reported experimental results. These results indicate that, in addition to sound predictions, the ML models can also observe complex microscale features and correlate them with macroscale device operation that can guide future experimentation for better results.

## 1. Introduction

Perovskite solar cells (PSCs) are the new inclusion that has become one of the most promising next-generation photovoltaic (PV) technologies with a power conversion efficiency (PCE) of 26.95% within a short period since their initial application [1,2]. Even with these impressive advances, optimization of PSC performance is often a challenging task because of the huge chemical and structural design space, as device efficiency is sensitive to subtle changes in composition and processing. There is also the complexity of interaction between various charge transport layers (CTLs) and the $ABX_3$ perovskite (PVK) absorber [3]. Though effective in making incremental improvements, traditional, trial-and-error methods are time-consuming, resource-consuming, and in many cases, ineffective in adequately exploring the multidimensional parameter space by which device performance is governed. While physical models have also been applied to analyze other thin-film technologies [4], the aforementioned issues in PSCs still pose a challenge. An alternative route that can be very powerful is machine learning (ML) [5].

In the ML viewpoint, PSCs are a perfect case study: they contain compositional freedom on the A, B, and X positions of the PVK layer and device-level parameters such as charge transport mobilities and band alignment at interfaces. This rich level of tuning poses an opportunity as well as a challenge for prediction modeling. Initially, in 2019, Li et al. used a sample of 333 experimentally reported samples to forecast 4 important PV metrics: open-circuit voltage ($V_{oc}$), short-circuit current density ($J_{sc}$), fill factor (FF), and PCE [6]. They used PVK composition descriptors, as well as energy level gaps between the PVK layer and the CTLs. This feature selection is physically intuitive in the sense that device efficiency is highly influenced by the properties of the absorber and interface energy arrangements that make charge extraction optimal. In a more recent paper, Zhao et al. have used compositional information as well as the explicit conduction band minimum (CBM) and the valence band maximum (VBM) values of the absorber and CTLs [7]. With their ML models, they obtained impressive accuracy in predicting PCE, using a sample count of 349. Liu et al. in another work, manually curated 248 datapoints from literature to train ML algorithms predicting PSC performance [8]. Their work also

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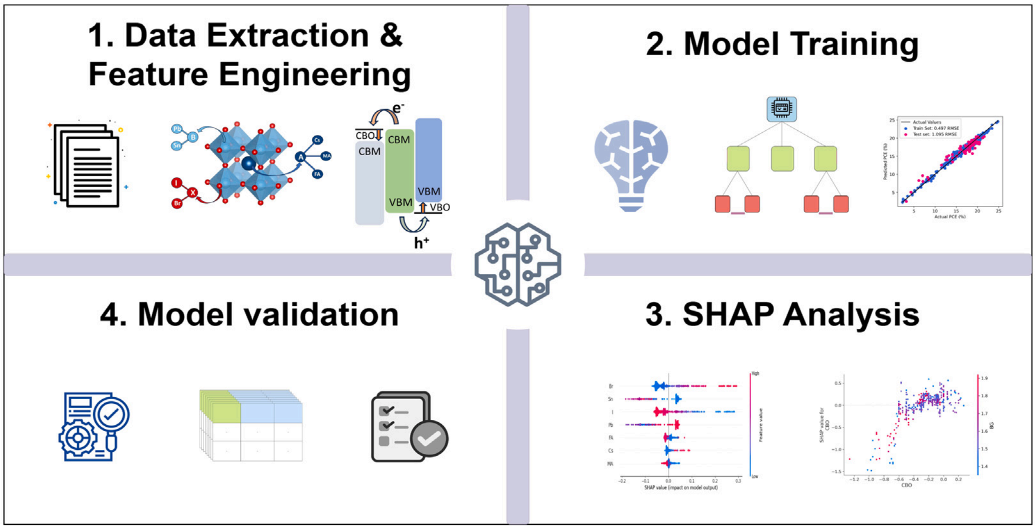


**Fig. 1.** Overall ML framework of our work.

incorporated the electron mobility of the ETL and the hole mobility of the HTL as input characteristics beyond compositional information and interfacial energy offsets, since they uniquely describe the charge transport characteristics of the ETL materials and the HTL material. In another study, the same group also scaled their ML architecture to make external quantum efficiency (EQE) predictions along with conventional PV quantities and further got more interpretable results through SHapley Additive exPlanations (SHAP) analysis to learn the best features [9]. ML has also been used in several other studies with alternative features of inputs in predicting PSC performance [10–14].

Although the applicability of ML in PV has been proven in previous studies, a number of limitations exist. Existing datasets are often limited in size, and the sets of features that were used previously are usually narrow, which often focus more on elementary composition features or separate energy levels, while ignoring features like carrier mobilities and architecture-dependent effects. The predictive performance is, therefore, limited. In addition, while some models achieve good predictive accuracy, they offer limited interpretability, making it difficult to derive meaningful physical insights that can guide device optimization. To fill such gaps, in our present work, we have manually curated a database of 503 records from various literature sources and used ML algorithms to predict pertinent PV properties using physically meaningful descriptors. To deepen the research on relative feature significance, the SHAP theory was used, which offers a physical understanding behind the model predictions [15]. Lastly, the trained models were compared with 12 separate PSC devices not used in the training or test set, to validate the predictive power on purely novel data.

## 2. Methodology

Fig. 1 reflects the general structure of our work. The adopted workflow in the present study involves four important steps that are to be used in order to achieve both predictive accuracy and interpretability. (1) Quality experimental data were extracted and curated using publications in peer-reviewed reports published after 2015. A feature engineering, in a while, was also conducted to produce useful descriptors. (2) The curated data were trained on four ensemble learning algorithms. The trained algorithms were then evaluated using 5-fold cross-validation, the root mean square error (RMSE), and the Pearson correlation coefficient ($r$) on both test and train sets. (3) SHAP-based interpretability analysis was used to identify the most influential features and their interactions with target variables. and (4) Finally, 12 independent samples were also used to test the trained algorithm. The framework ensures relevant data preparation and model training to be predictive and provide physically informative insights on the PSCs. These steps are more detailed in the following sub-sections.

### 2.1. Data extraction and feature engineering

The good performance of the ML model heavily relies on the quality and relevance of the training data. Thus, it is necessary to make sure that well-defined and correlated input features are present in the dataset to capture the underlying physical processes and enhance the predictive accuracy of the model. Thus, to train our ML models, we selected with care a total of 503 datapoints that have been gathered manually through peer-reviewed research articles on the subject that have been published not older than 2015. This dataset contains device architectures of both types, nip and pin; however, only device architectures with a single layer were considered. Absorber compositions were limited to Cs, FA, and MA in the A-site; Pb and Sn in the B-site, and I/Br in the X-site. We have also attempted to be consistent by only incorporating the reports that clearly gave the critical details regarding the PVK composition (Cs, FA, MA, Pb, Sn, I, Br), PVK conduction band minimum (CBM), PVK valence band maximum (VBM), conduction band minimum (ETL_CBM) of electron transport layer (ETL), valence band maximum (HTL_VBM) of hole transport layer (HTL), and the charge carrier mobilities (e_mobility and h_mobility). In this case, the e_mobility and the h_mobility represent the mobility of the electrons in ETL and the mobility of holes in HTL, respectively. In cases where some of these feature parameters were not mentioned in the original articles (which affects a small percentage of the overall extracted information, which is about 2%), we used an imputation strategy followed by Liu et al. to preserve the dataset size [8,9]. Specifically, in cases of CTLs, where the energy levels or carrier mobilities were not measured in the particular study, we used standard values found in the commonly used literature for that specific material. The complete list of these reference values used for imputation is provided in Table S3 of Supplementary File 1 (S1). This imputation methodology ensures that high-quality data records are not lost due to the absence of a few features. Besides that, we used one-hot encoding to select the device configuration, thus adding two additional columns, arch_nip and arch_pin, which represent the nip and pin architectures.

In addition to direct features extraction, two other features conduction band offset (CBO), and the valence band offset (VBO) were also determined by means of Eqs. (1) and (2)[16]. These characteristics

**Table 1**
Description of dataset features used for training ML models.

| Feature | Type | Description |
|---|---|---|
| PCE | Target | Power conversion efficiency (%) |
| $V_{oc}$ | Target | Open-circuit voltage (V) |
| $J_{sc}$ | Target | Short-circuit current density (mA/cm$^2$) |
| FF | Target | Fill factor (dimensionless) |
| Cs, FA, MA | Composition | A-site cation fractions: Cesium, Formamidinium, Methylammonium |
| Pb, Sn | Composition | B-site cations: Lead and Tin fractions |
| I, Br | Composition | X-site anions: Iodine and Bromine fractions |
| CBO | Band offset | Conduction band offset between absorber and ETL (eV) |
| VBO | Band offset | Valence band offset between absorber and HTL (eV) |
| CBM | Energy level | Conduction band minimum of absorber (eV) |
| VBM | Energy level | Valence band maximum of absorber (eV) |
| ETL_CBM | Energy level | Conduction band minimum of ETL (eV) |
| HTL_VBM | Energy level | Valence band maximum of HTL (eV) |
| e_mobility | Transport property | Electron mobility of ETL (cm$^2$/Vs) |
| h_mobility | Transport property | Hole mobility of HTL (cm$^2$/Vs) |
| `arch_nip` | Architecture | Binary indicator (1/0) for nip device structure |
| `arch_pin` | Architecture | Binary indicator (1/0) for pin device structure |

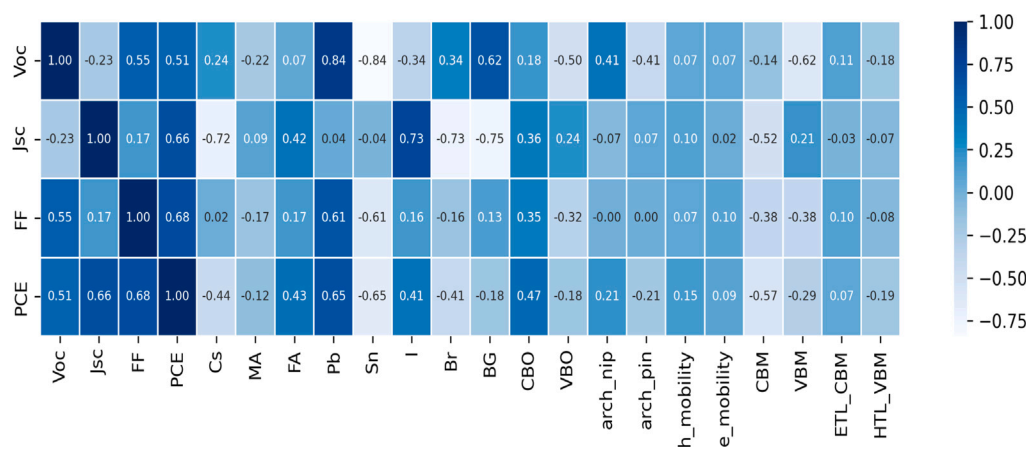


**Fig. 2.** Correlation heatmap of target features with input features in D1.

measure the band alignment of the PVK absorber and underlying CTLs, which gives significance to the interfacial charge transfer mechanism.

$$\mathrm{CBO} = E_{\mathrm{CBM,ETL}} - E_{\mathrm{CBM,Perovskite}} \tag{1}$$

$$\mathrm{VBO} = E_{\mathrm{VBM,Perovskite}} - E_{\mathrm{VBM,HTL}} \tag{2}$$

The full final dataset (D1) can be found in the supplementary file (S2). This dataset comprises a total of 503 records with 21 well-defined columns. Of these, four columns correspond to the target variables: PCE, $V_{oc}$, $J_{sc}$, and FF, while the remaining columns represent the input features used to train the algorithms. These include compositional descriptors (Cs, FA, MA, Pb, Sn, I, Br), engineered features (CBO and VBO), electronic energy levels (CBM, VBM, ETL_CBM, and HTL_VBM), charge transport properties (e_mobility and h_mobility), and device architectural identifiers (`arch_nip` and `arch_pin`). Table 1 provides a detailed description of each feature.

According to the correlation heatmap in Fig. 2, we can see that all the target variables in the given study are characterized by significant correlations with several input features in the dataset. This indicates that the chosen features can be useful to forecast device performance and that there exist significant correlations between material features, interfacial conditions, and photovoltaic results. In addition, a full correlation heatmap of the extracted input features can be seen in Figure S1 of the supplementary file 1 (S1), which gives a more detailed view of the inter-feature correlations.

### *2.2. ML algorithm and model evaluation*

It is vital to implement an ML model that is both accurate and explainable to achieve model reliability and to obtain information that has physical meaning. We utilized four common ensemble-based regression algorithms: Random Forest (RF) [17], Gradient Boosting Regressor (GBR) [18], Extreme Gradient Boosting (XGB) [19], and CatBoost (CB) [20]. These algorithms were implemented using Scikit-learn [21], XGBoost, and CatBoost libraries. The ensemble approaches in particular are better adapted to this operation, typically making use of an aggregate of many base learners, with the aim of increasing generalization and reducing overfitting to achieve improved predictive accuracy [22]. Further, these algorithms are interpretable and very useful in explaining the physical feature contribution parameters that affect PSC performance. Although other potent ML models like artificial neural networks (ANN) and support vector machines (SVM) have been broadly applied in materials science, we have prioritized ensemble tree-based algorithms in this study. The rationale was two-fold: First, ensemble approaches have been experimentally observed to do better on small-to-medium-sized tabular datasets, including about 500 observations, whereas neural networks are more feasible when large amounts of data are present [23]. Second, tree-based model types are also more compatible with post-hoc interpretability methods such as SHAP, which means that one can more clearly apply tree-based models to extract physical structure–property interactions than relative black-box alternatives such as SVM or ANN [23].

Two complementary measures of evaluation, the RMSE and $r$, were then used to quantitatively evaluate the predictive performance of each model. As shown in Eq. (3), RMSE is used to determine how large the standard deviation of the prediction error values is, i.e., the average size of error between prediction and observed value. Minimized RMSE means a better model. The $r$ value, as calculated by Eq. (4), determines how large and in what direction actual measurements align with predictions. A near value of $r$ to 1 indicates greater correlation.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

$$r = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}\sqrt{\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}} \tag{4}$$

Here, $y_i$ and $\hat{y}_i$ represent the actual and predicted values, respectively, for the $i$th data point, $n$ denotes the total number of observations, $\bar{y}$ is the mean of the observed values, and $\bar{\hat{y}}$ is the mean of the predictions. Together, these metrics provide us with an adequate evaluation of the model performance since they evaluate the accuracy of the predictions and the correlation to actual values.

### 2.3. Model interpretation

It is as essential to interpret the decisions of the ML models as it is to reach high predictive accuracy, especially when it comes to using ML in the sphere of science, where interpretations of underlying relationships are vital. Using the parameters that contribute most significantly to the performance of the devices not only ensures that the model is validated, but also helps design further experiments. To reveal the relative importance of each feature in predicting the performance metrics of PSCs, in this work, we used the SHAP method [15]. The reasoning behind SHAP values is based on cooperative game theory, where individual features are treated as players that contribute to the output prediction. They quantify the marginal contribution of each aspect to a specific prediction, making it globally and locally interpretable. Mathematically, a prediction equation of the model of an instance $x$ can be given by:

$$f(x) = \phi_0 + \sum_{i=1}^{M}\phi_i \tag{5}$$

where $f(x)$ is the model output, $\phi_0$ is the base value (mean prediction across the training dataset), $M$ is the total number of features, and $\phi_i$ is the SHAP value of feature $i$, representing the contribution that feature i makes to the deviation in the prediction with respect to the base value. The positive SHAP value means that the feature shifts the prediction above the model baseline value (by enhancing the predicted performance measure), and the negative SHAP value indicates that a feature reduces the prediction in comparison with the model baseline. This interpretation makes it possible to see how specific factors, such as specific perovskite composition or specific initial energy levels, affect the prediction of photovoltaic performance. SHAP value of feature $i$ is the weighted marginal contribution of the feature to all possible subsets of features and given as:

$$\phi_i = \sum_{S\subseteq F\setminus\{i\}} \frac{|S|!(M-|S|-1)!}{M!}\left[f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)\right] \tag{6}$$

where $F$ is the full feature set, $S$ is a subset of features excluding $i$, and $f_S(x_S)$ is the prediction of the model based on the features contained in $S$. The formulation ensures that SHAP quantifies the contribution in the same proportion to all potential feature arrangements.

## 3. Results

### 3.1. Performance prediction

On all four PV performance metrics, four ensemble learning models- RF, GBR, XGB, and CB- were trained. This provided 16 trained models altogether. To simplify, each model was labeled by a combination of the algorithm and the desired metric, e.g., $RF_{PCE}$ represents the RF model applied in PCE prediction, $GBR_{FF}$ represents the GBR model in FF prediction, etc. A fixed random state was used to divide the dataset D1 into training and testing sets of an 85:15 ratio to guarantee reproducibility. Systematic evaluation was then performed using a 5-fold cross-validation (CV) when the models were trained using an 85% train set. In a 5-fold CV, the training dataset is divided into five equal subsets. The model is trained on four subsets and validated on the remaining one subset. This process is repeated five times, with every subset used at least once as the validation set. This would avoid the risk of biased assessment and give a more accurate estimation of the overall generalization of the model as it minimizes variance due to a certain train-test split [24]. The highest performing model was able to get an average RMSE of 1.82%, 0.0552 V, 1.365 mA/cm$^2$, and 0.0418 for PCE, $V_{OC}$, $J_{SC}$, and FF, respectively, during 5-fold CV. The corresponding standard deviations were 0.302% for PCE, 0.0042 V for $V_{OC}$, 0.1654 mA/cm$^2$ for $J_{SC}$, and 0.0042 for FF, which revealed the stability of the CV results.

**Table 2**
Best-performing algorithms with their RMSE and $r$-values on training and testing sets.

| Algorithm | Training set | | Testing set | |
|---|---|---|---|---|
| | RMSE | $r$-value | RMSE | $r$-value |
| $CB_{PCE}$ | 0.497% | 0.991 | 1.095% | 0.964 |
| $RF_{Voc}$ | 0.022 V | 0.991 | 0.033 V | 0.979 |
| $RF_{Jsc}$ | 0.543 mA/cm$^2$ | 0.988 | 1.013 mA/cm$^2$ | 0.958 |
| $RF_{FF}$ | 0.016 | 0.978 | 0.031 | 0.614 |

All models, on 15% of data points that are withheld as the test set, were then evaluated to determine predictive accuracy. The RMSE and the $r$-values obtained with all 16 models corresponding to PCE, $V_{oc}$, $J_{sc}$, and FF predictions, respectively, are summarized in Table S1 of the S1. The most accurate of the algorithms in PCE prediction was CB, with an RMSE of 1.095%, and an $r$-value of 0.924, demonstrating a strong linear relationship between predicted and experimental results. Likewise, RF models were the most efficient in predicting all other remaining parameters- $V_{oc}$, $J_{sc}$, and FF. Table 2 contains the RMSE and $r$-values in detail on the training and testing sets of these best-performing models. In general, the chosen ensemble algorithms performed very well in predicting accuracy, and CB showed better predictive abilities at the PCE prediction, and RF showed maximized prediction for $V_{oc}$, $J_{sc}$, and FF.

In comparison to PCE, $V_{oc}$, and $J_{sc}$, the predictive performance for the FF was noticeably lower ($r = 0.614$ on the test set). This behavior is physically reasonable because FF is a composite parameter that is very sensitive to various non-ideal loss mechanisms, including bulk/interface recombination, series and shunt resistance, and hysteresis-related ion migration [25,26]. Many of these device-level effects are not explicitly represented in our current feature set. Owing to the unavailability of such parameters, the model is unable to fully capture the multivariate physical processes governing FF. Thus, under the current feature description, FF prediction remains intrinsically more challenging than the other photovoltaic parameters.

Fig. 3 displays the overall fitting results of the best models, $CB_{PCE}$, $RF_{Voc}$, $RF_{Jsc}$, and $RF_{FF}$, on the training and test data sets. The close clustering and the narrow spread of scatter points along the black diagonal line show the good accuracy between the predicted and true values, which reinforces the reliability of these models. The fitting results for all other models are provided in Figures S2–S5 of the S1, which further demonstrate the reasonable predictive performance of the full set of algorithms.

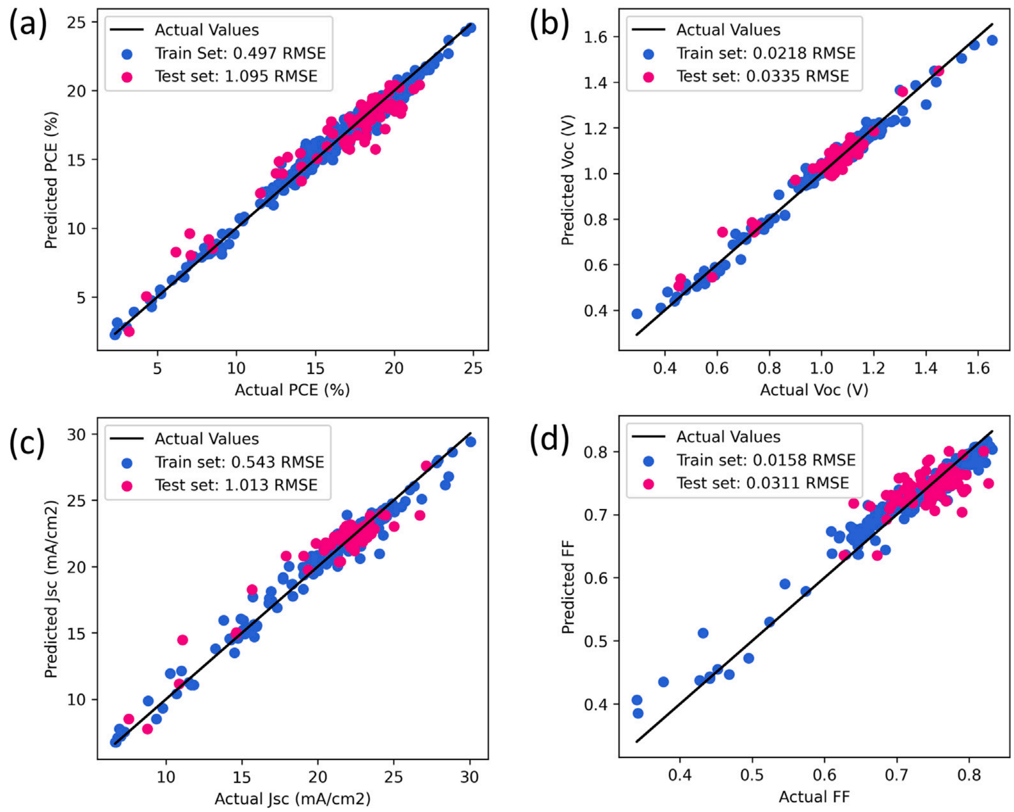


**Fig. 3.** Fitting performance of the best-performing models on training and testing sets: (a) $CB_{PCE}$, (b) $RF_{Voc}$, (c) $RF_{Jsc}$, and (d) $RF_{FF}$.

### 3.2. SHAP analysis

Fig. 4 shows the SHAP beeswarm plot, which indicates the seven most dominant features behind the predictive choices of our models. The entire beeswarm plot of all the input features is listed in Figure S6 of the S1. The high feature values in this visualization are represented by red points, and the low feature values by blue points, and various displacements on the horizontal line depict the magnitude of their contribution to the prediction, respectively. Just as observed by Liu et al. in their study [9], one key observation from the beeswarm plots is that a significant portion of the influential features is directly correlated to the PVK absorber itself, confirming the idea that the inherent material characteristics of the PVK layer determine the PV performance in PSCs. Of the A-site cations, the proportion of FA seems especially significant, producing a constantly positive effect on all four performance indicators. This correlates with experimental results, in which FA addition is experimentally known to stabilize the PVK structure and enhance the device performance [27]. It is also a well-established fact that the bandgap of the $ABX_3$ perovskites is highly dependent on the $BX_6$ octahedral structure and that the bandgap can be systematically tuned by changing the composition at the B site (Pb/Sn) or the X site (I/Br) [28]. It therefore follows that performance parameters such as $J_{sc}$ and $V_{oc}$, which are tightly coupled to the absorber bandgap, are found through SHAP analysis to be most strongly influenced by B- or X-site features. The importance assigned by the model to these descriptors suggests that it has effectively captured the microscale physics governing bandgap variation and, in turn, the photovoltaic characteristics of the device. The hole mobility ($h$_mobility) of the HTL is another characteristic of central significance (Fig. 4). The SHAP distribution shows that increased h_mobility exerts a consistently positive influence on all metrics of device performance, indicating its vital role in supporting efficient charge extraction and reducing recombination losses.

To explore the effects of individual features in even more detail, we can use SHAP dependence plots of selected important features, which show the effects of a change in the value of a single feature within the context of the model predictions. Fig. 5(a–c) shows the dependence of the A-site cations (FA, MA, and Cs) on PCE. In the case of FA content (Fig. 5a), the SHAP values are negative when the concentration of FA is lower than 0.5. This implies that the model relates decreased FA fractions with worsened device performance. The FA concentration, however, when it becomes greater than 0.5, the SHAP values start rising in a pronounced fashion, and the high SHAP values are seen after the concentration is above 0.9. This pattern indicates that the model strongly correlates high FA content with high PCE. This association aligns with experimental knowledge that high FA content elevates the degree of crystallinity, thermal stability, and optoelectronic properties of perovskites [27]. The dependence plots of MA and Cs Fig. 5b and c, on the other hand, give a different trend. The optimal SHAP values are observed at lower concentration levels of both cations (around 0.05), thus indicating that the model identifies a small amount of MA or Cs with high FA concentration (inferred by the red plots) can positively influence the performance of the devices. This trend is consistent with the success of the triple-cation configurations reported in the literature, where small amounts of MA and Cs are employed to stabilize the photoactive FA phase rather than serving as the primary cation [29,30]. Their SHAP values, however, fall quickly for their higher concentrations, suggesting that the model outputs lower PCE for compositions deviating from this small concentration interval. From this analysis, it can be seen that the model predicts the most effective A-site way of forming the very effective device has a prevailing FA fraction in the range of 0.9–0.95, and a slight concentration of MA or Cs [29].

As shown in Fig. 5d, the model predicts that Br-additions up to a concentration of 0.6 (of the total X-site fraction of 3) are related to small, yet positive SHAP values. Nevertheless, there is a sharp decrease in SHAP contribution after this threshold. This indicates that bandgap tuning for PVK with the Br concentration (0–0.6) has comparatively

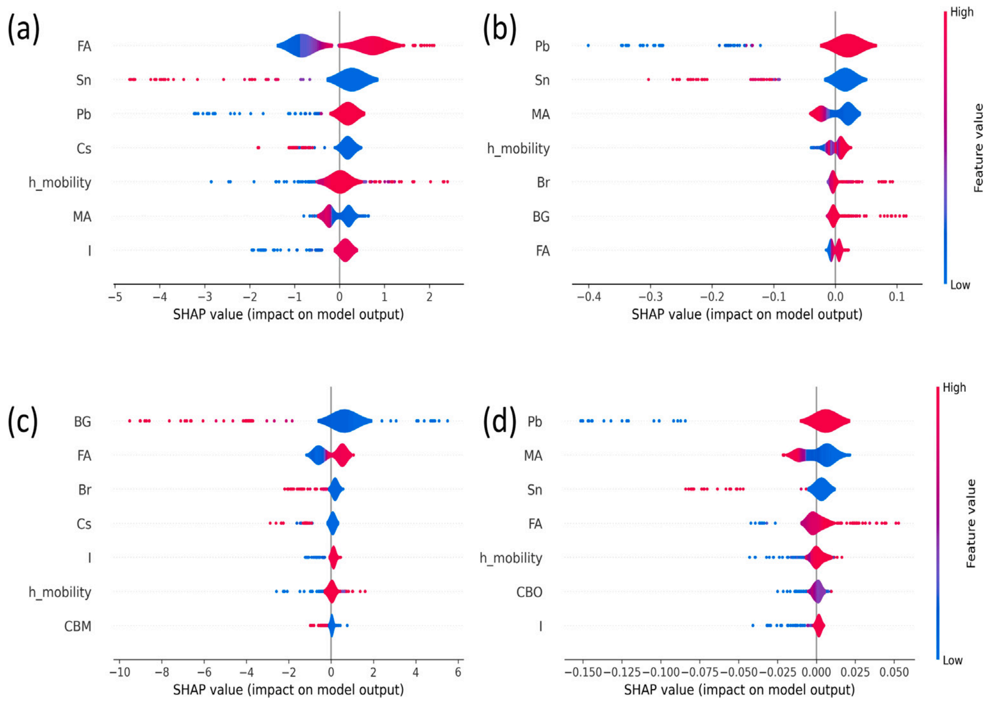


**Fig. 4.** SHAP Beeswarm plot of top 7 features for (a) $CB_{PCE}$, (b) $RF_{Voc}$, (c) $RF_{Jsc}$, and (d) $RF_{FF}$.

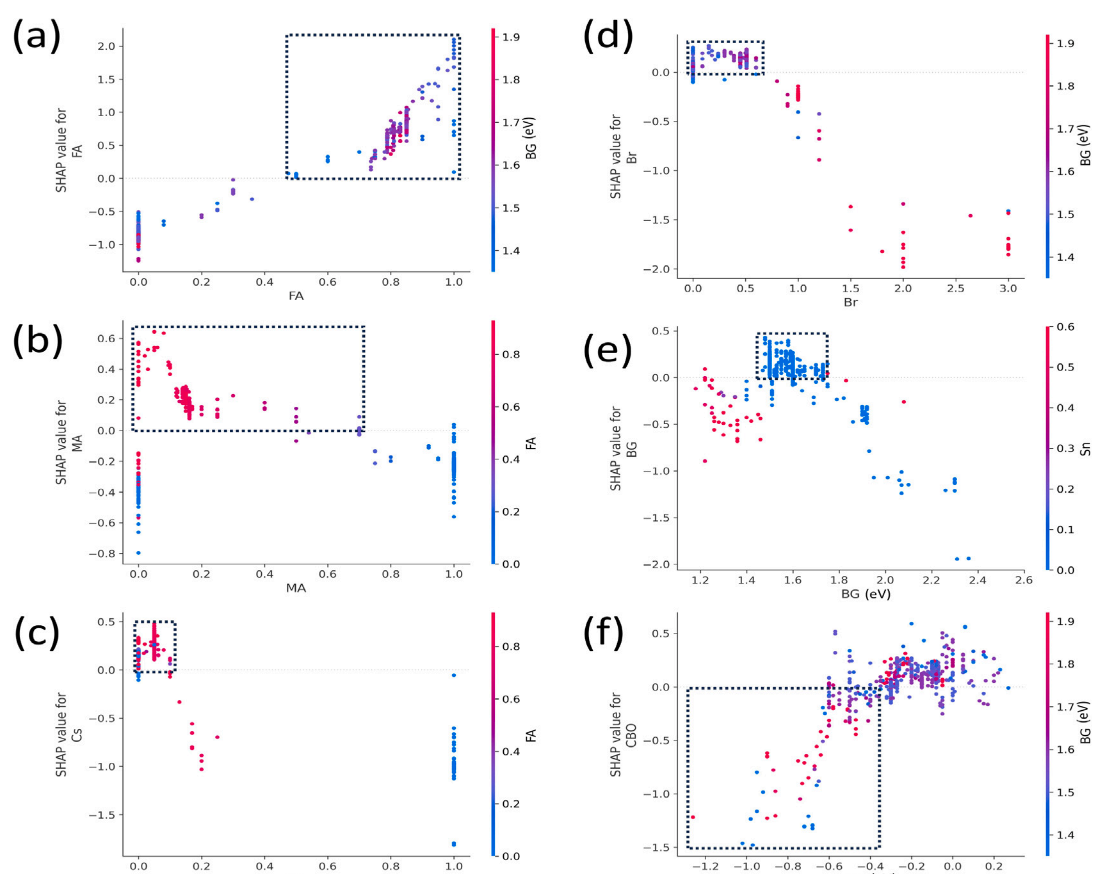


**Fig. 5.** Feature dependence plots of some influential parameters for PCE prediction: (a) FA, (b) MA, (c) Cs, (d) Br, (e) Bandgap (BG), and (f) CBO.

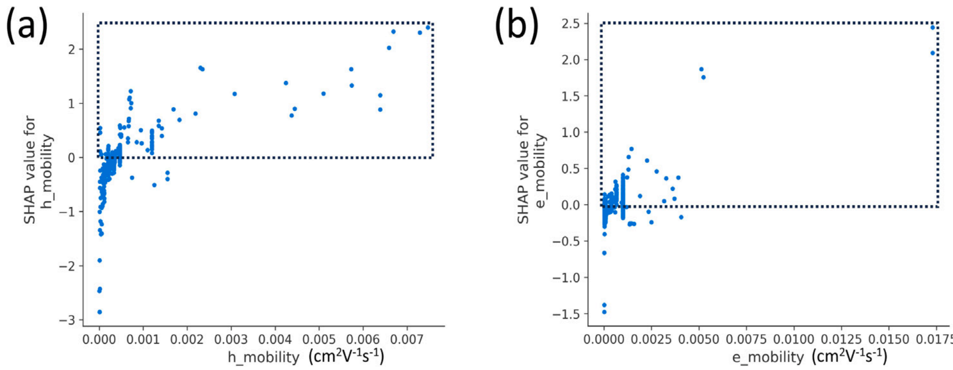


**Fig. 6.** Feature dependance plots for PCE prediction: (a) h_mobility, and (b) e_mobility.

**Table 3**
Comparison of experimental and ML predicted parameters of PSCs reported in different literatures.

| Device | Arch. | PCE | ML PCE | Voc | ML Voc | Jsc | ML Jsc | FF | ML FF | Ref. |
|---|---|---|---|---|---|---|---|---|---|---|
| ITO/W-TiO2/$Cs_{0.05}FA_{0.81}MA_{0.14}PbI_{2.55}Br_{0.45}$/Spiro-OMeTAD/Au | nip | 20.59 | 20.51 | 1.16 | 1.16 | 22.56 | 22.52 | 0.780 | 0.778 | [32] |
| ITO/PTAA/$Cs_{0.05}MA_{0.1425}FA_{0.8075}PbI_{2.55}Br_{0.45}$/$PC_{61}BM$/Ag | pin | 19.30 | 18.49 | 1.10 | 1.089 | 22.70 | 22.63 | 0.778 | 0.756 | [33] |
| FTO/NiOx/$Cs_{0.17}FA_{0.83}PbI_{2.4}Br_{0.6}$/$C_{60}$/BCP/Cu | pin | 16.14 | 16.56 | 1.06 | 1.09 | 19.02 | 20.52 | 0.800 | 0.745 | [34] |
| ITO/PEDOT:PSS/$MAPbI_3$/$PC_{61}BM$+BCP/Ag | pin | 14.40 | 14.03 | 0.92 | 0.949 | 20.30 | 19.93 | 0.770 | 0.749 | [35] |
| FTO/meso-Zn-$TiO_2$/$MAPbI_3$/Spiro-OMeTAD/Ag | nip | 16.80 | 16.65 | 1.05 | 1.083 | 21.60 | 20.90 | 0.745 | 0.749 | [36] |
| FTO/c-$TiO_2$/$Cs_{0.05}MA_{0.12}FA_{0.83}PbI_{2.55}Br_{0.45}$/Spiro-OMeTAD/Au | nip | 19.14 | 19.62 | 1.12 | 1.112 | 23.00 | 22.87 | 0.743 | 0.749 | [37] |
| ITO/PEDOT:PSS/$MAPbI_3$/$PC_{61}BM$/Al | pin | 15.74 | 16.40 | 0.91 | 0.992 | 21.10 | 21.13 | 0.820 | 0.758 | [38] |
| FTO/c-$TiO_2$/$CsPbIBr_2$/Spiro-OMeTAD/Ag | nip | 9.06 | 8.81 | 1.20 | 1.213 | 11.60 | 10.97 | 0.648 | 0.703 | [39] |
| FTO/$TiO_2$/$FA_xMA_{1-x}PbI_{3-y}Br_y$/Spiro-OMeTAD/Ag | nip | 16.74 | 17.19 | 1.09 | 1.079 | 20.05 | 22.66 | 0.730 | 0.748 | [40] |
| ITO/TPA-NAP-TPA/$MAPbI_3$/$PC_{60}BM$/BCP | pin | 14.63 | 14.72 | 0.96 | 1.011 | 21.55 | 20.62 | 0.710 | 0.721 | [41] |
| ITO/TPA-TVT-TPA/$MAPbI_3$/$PC_{60}BM$/BCP | pin | 16.32 | 14.96 | 1.07 | 1.008 | 21.49 | 20.77 | 0.710 | 0.708 | [41] |
| FTO/$TiO_2$/$Cs_{0.05}FA_{0.81}MA_{0.14}PbI_{2.55}Br_{0.45}$/Spiro-OMeTAD | nip | 18.26 | 18.32 | 1.044 | 1.093 | 23.60 | 22.97 | 0.741 | 0.738 | [42] |

consistent PCE values, but with Br content (0.6 to 3) shows a negative influence, resulting in lower PCE. Furthermore, values of BG within the 1.5–1.7 eV range show a positive contribution to PCE, as in Fig. 5e. Beyond this range, especially with larger band gaps ($BG > 1.7$ eV), it can be seen that PCE contribution diminishes clearly. Nevertheless, at the narrow BG ($BG < 1.4$ eV), the smaller SHAP values may be explained by the increased Sn concentration, as depicted by red plots in Fig. 5e. In this range, the data points disperse vertically based on Sn concentration: samples with high Sn content (red points) exhibit significantly lower SHAP values compared to low-Sn samples at the same BG. This can be attributed to the reality that excess Sn content usually lowers device PCE, as also observed in the beeswarm plot (Fig. 4a). The SHAP dependence plot of the CBO is shown in Fig. 5(f). The analysis demonstrates that CBO values that are strongly negative (−0.4 eV and below) are assigned significantly negative SHAP values by the model, which resembles the formation of an unfavorable cliff-like Band structure that enhances interfacial recombination [31]. Conversely, CBO values in the vicinity of approximately −0.3 eV to +0.2 eV have a favorable SHAP value, as it is the ideal case of near-flat band alignment, which helps the efficient electron transport process.

Another critical parameter shown by Fig. 6(a) is h_mobility within the HTL, where the SHAP values lie within −3 to +2. Devices with very low h_mobility exhibit highly negative influences on PCE. A clear trend is observed where rising mobility correlates with increasingly positive SHAP contributions. The same applies to e_mobility in ETL, as seen in Fig. 6(b). Nonetheless, in addition to high carrier mobilities, it is also crucial to make sure that the energy levels of CTLs align well with the absorber energy levels. Figure S10 in the Supplementary Information S1 presents feature dependence plots of some major features on the remaining photovoltaic parameters. Collectively, these FD plots support the fact that the best photovoltaic performance involves well-tuned features. It must also be noted that although SHAP analysis can clarify the feature contributions made by the model, these associations are based on the statistical correlations that appear in the training data, but are not straight, isolated causality.

### 3.3. Model validation and comparison

Finally, in order to critically compare the generalizability of our ML framework, we went further by testing it using a novel dataset of 12 samples. These samples were made to reflect data points not introduced in the initial training sample or the testing set. Interestingly, these predictions had a good agreement with the experimentally reported values, which are summarized in Table 3. As observed, both nip and pin architectures with varying PVK composition and different CTLs are found in the samples. Table S2 of the S1 lists the detailed input features relative to these 12 samples. Such a level of compatibility between the predicted and experimental values proves the predictiveness of the trained ML models.

Upon comparison with previously published studies on ML-based prediction of PV performance metrics, our models demonstrate a clear improvement in prediction accuracy. A detailed comparison of our results with earlier works is provided in Table 4. In particular, for PCE prediction, we achieved a minimum RMSE of 1.09%, which is the lowest value reported in the literature. Similarly, the RMSE values obtained for $V_{oc}$, $J_{sc}$, and FF were 0.033 V, 1.03 mA/cm$^2$, and 0.02, respectively, representing improved results. The $r$-value of all the metrics is also consistently improved. Altogether, this demonstrates that this work offers an enhancement in the existing literature.

## 4. Conclusion

In the present work, we have proposed a complete ML-based approach aimed at predicting the most significant photovoltaic parameters of PSCs, including the $V_{oc}$, $J_{sc}$, FF, and PCE. Using a highly filtered database of 503 datapoints based on experimental reports, we trained ML models based on physically relevant features, such as PVK compositions, CBO, VBO, energy levels, and charge carrier mobilities. Our models had outstanding accuracy, with CatBoost showing the best results on PCE and RF providing the best results on $V_{oc}$, $J_{sc}$, and FF.

**Table 4**
Performance comparison of our ML models with prior similar literature work.

| References | Year | Dataset size | RMSE | | | | r-value | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | PCE (%) | $V_{oc}$ (V) | $J_{sc}$ (mA/cm$^2$) | FF | PCE | $V_{oc}$ | $J_{sc}$ | FF |
| Li et al. [6] | 2019 | 333 | 3.23 | 0.085 | 3.200 | 0.061 | 0.80 | 0.800 | 0.820 | 0.610 |
| Liu et al. [8] | 2022 | 248 | 1.58 | 0.051 | 1.040 | 0.046 | 0.86 | 0.930 | 0.960 | 0.630 |
| Liu et al. [9] | 2024 | 418 | 1.25 | 0.049 | 1.430 | 0.032 | 0.94 | 0.917 | 0.920 | 0.810 |
| Hui et al. [11] | 2024 | 615 | 2.63 | 0.078 | 2.244 | 0.057 | 0.74 | 0.865 | 0.824 | 0.391 |
| Aggarwal et al. [12] | 2024 | 447 | 2.84 | 0.100 | 2.706 | N/A[a] | 0.86 | 0.920 | 0.910 | N/A[a] |
| This work | 2025 | 503 | 1.09 | 0.033 | 1.013 | 0.031 | 0.96 | 0.979 | 0.958 | 0.614 |

[a] N/A – Not available, as the corresponding metric values were not reported in this study.

The strength of the trained models was also illustrated by the benchmarking capability on 12 additional experimental samples, not covered during the training phase, whose values gave good correspondence with the known values. In comparison to the previous literature, our models consistently presented higher predictive accuracy of all measures of performance. SHAP analysis offered an explainable perspective of the decision-making process, with intrinsic perovskite properties, HTL hole mobility, and band alignment at the interfaces proving to be the most crucial factor to the device efficiency. The implications of those insights are that they not only represent physical consistency of the ML framework but also provide future design considerations of materials and devices. Altogether, the proposed work reveals promising opportunities for ML as a potent adjunct of experimental/computational methods to the quick, precise, and interpretable prediction for PSCs growth. Although the research provides a strong predictive model for single-junction PSC devices, we acknowledge that there are some limitations posed by the existing dataset. Specifically, the current feature set is limited to standard A-site (Cs, FA, MA), B-site (Pb, Sn), and X-site (I, Br) compositions. Future efforts should seek to broaden this compositional space with additional elements mixing (such as Rubidium (Rb), Germanium (Ge), and Chlorine (Cl)), which are more useful in BG tuning as well as in stability enhancement. Moreover, in the current work, single-layer absorbers are considered, which can be extended to incorporate tandem designs, which can be useful to capture more complex device physics in such architectures.

## CRediT authorship contribution statement

**Subham Subba:** Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Suman Chatterjee:** Writing – review & editing, Validation, Supervision, Formal analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.solmat.2026.114215.

## Data availability

The data and code essential for conducting the analyses of this study are available at this https://github.com/subhamsubba/PCE GitHub link.

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